

Comprehensive Engineering Audit Report

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1. Executive Summary & Operational Assessment

This audit validates the closed-loop performance of the Hierarchical Adaptive Fuzzy Irrigation Control System operating under the **Normal** scenario for a 24-hour simulation duration. Total water applied across all 3 zones was **5760.0 Liters**. Physical mass balance was strictly maintained with a peak residual of **0.000000 mm**.

Agronomic AI Advisor Diagnostic:

****Telemetry & Fuzzy Control Diagnostic for Simulation Scenario 'Normal'**:**

- ****Total Water Applied**:** 5760.0 Liters across all zones.
- ****Mass Balance Conservation**:** Maximum residual was 0.000000 mm (strict physical conservation maintained).

****Zone-by-Zone Operational Analysis**:**

- ****Zone 1 (Tomato in Loam)**:** Received 0.0 L. Maintained tracking MAE of 0.00%. Minimum soil moisture reached 55.0%. The MainIrrigationFIS modulated pulses whenever moisture dropped below target, balancing crop evapotranspiration against leaching.
- ****Zone 2 (Wheat in Sandy)**:** Received 0.0 L. Maintained tracking MAE of 0.00%. Minimum soil moisture reached 42.0%. The MainIrrigationFIS modulated pulses whenever moisture dropped below target, balancing crop evapotranspiration against leaching.
- ****Zone 3 (Maize in Clay)**:** Received 0.0 L. Maintained tracking MAE of 0.00%. Minimum soil moisture reached 65.0%. The MainIrrigationFIS modulated pulses whenever moisture dropped below target, balancing crop evapotranspiration against leaching.

***Note:** Under shared supply constraints, Layer C Bounded Priority-Weighted Allocation prioritizes higher-demand and higher-priority zones without exceeding available reservoir volumes.*

2. System & Zone Physical Configurations

Zone ID	Crop Type	Soil Type	Area (m²)	Target SM (%)	FC (%)	WP (%)	Priority
1	Tomato	Loam	100.0	68.0%	70.0%	25.0%	80%
2	Wheat	Sandy	120.0	55.0%	60.0%	18.0%	25%
3	Maize	Clay	80.0	65.0%	75.0%	30.0%	75%

3. Closed-Loop Performance Metrics

Zone	Applied (L)	Applied (mm)	MAE (%)	Fulfillment (%)	Mean SM (%)	Max SM (%)	Deficit (h)
Zone 1	3804.8	38.05	10.74%	42.4%	57.3%	—	0.0h
Zone 2	1240.0	10.33	12.71%	6.0%	42.3%	—	0.0h
Zone 3	715.2	8.94	0.11%	57.4%	65.1%	—	0.0h

4. Physical Conservation & Water Balance Audit

Conservation Law: $\Delta\text{Storage} = \text{Infiltration} + \text{Irrigation} - \text{ETc} - \text{Deep Percolation} - \text{Runoff}$

- **Simulation Max Residual:** 0.00000000 mm
- **Audit Status:** **100% PASS** (Residual < 1e-4 mm threshold)

- **Supervisory Allocation Invariant:** $\text{Sum}(\text{Allocated_z}) \leq \text{Available_Supply}$ (Enforced by Layer C iterative capped water-filling).
- Conservation Law:** $\Delta\text{Storage} = \text{Infiltration} + \text{Irrigation} - \text{ETc} - \text{Deep Percolation} - \text{Runoff}$
- **Simulation Max Residual:** 0.00000000 mm
 - **Audit Status:** **100% PASS** (Residual < 1e-4 mm threshold)
 - **Supervisory Allocation Invariant:** $\text{Sum}(\text{Allocated_z}) \leq \text{Available_Supply}$ (Enforced by Layer C iterative capped water-filling).

5. Offline PSO Parameter Tuning Summary

The supervisory actuator controller (MainIrrigationFIS) has been systematically calibrated via continuous Particle Swarm Optimization across 18 membership function transition breakpoints:

- **Baseline Composite Fitness J:** 0.16521
- **Optimized Composite Fitness J*:** 0.13620 (-15.4% Cost Reduction)
- **Average Water Savings:** 8.7% volumetric reduction
- **Control Smoothness:** -28.5% valve chatter mitigation
- **Safety Invariant:** Strict offline execution; runtime control remains deterministic Mamdani inference.